

Notes: Sah and Stiglitz (AER, 1986)

The Architecture of Economic Systems: Hierarchies and Polyarchies

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1 Big picture

This note summarizes Sah and Stiglitz (1986), which develops a simple theoretical framework for comparing different organizational architectures when individual decision makers make mistakes. The paper studies how **hierarchies** and **polyarchies** aggregate imperfect judgments, and shows that the same screening technology can generate different system-level errors depending only on the organization of decision authority.

The central idea is that economic systems are not merely collections of agents; they are **architectures of decision making**. Architecture specifies who screens projects, who has authority to accept or reject, and how information is transmitted. Because individuals sometimes reject good projects and accept bad projects, organizational design determines how these individual errors are filtered or amplified.

1.1 One-sentence summary

The paper builds a model in which imperfect project screening generates Type-I and Type-II errors, and shows that **polyarchies** tend to accept more good and bad projects by giving rejected projects a second chance, whereas **hierarchies** tend to reject more good and bad projects by requiring sequential approval.

1.2 Key question

When decision makers are imperfect, which architecture performs better: a decentralized polyarchy, where multiple decision makers can independently accept projects, or a centralized hierarchy, where projects must pass through sequential screens?

1.3 Core answer

There is no universally superior architecture. A polyarchy is better when it is more costly to reject good projects; a hierarchy is better when it is more costly to accept bad projects. The optimal architecture depends on the quality of the project pool, the accuracy and tightness of screening, the cost of communication, and the relative losses from the two types of errors.

2 Incremental contribution of the paper

The paper's incremental contribution is to introduce an explicit analytical language for comparing economic systems as **decision architectures**. Rather than asking whether markets or bureaucracies are better in a broad ideological sense, Sah and Stiglitz isolate a precise mechanism: how imperfect local judgments become system-level outcomes.

2.1 Contribution relative to standard market-versus-bureaucracy debates

Earlier discussions of market economies and centralized economies often emphasized incentives, prices, planning algorithms, or resource allocation. This paper adds a different channel: **error aggregation**. A market-like system may perform better not only because of prices or incentives, but also because a good idea rejected by one firm may still be accepted by another. A bureaucracy-like system may perform better not only because of coordination, but also because sequential review can prevent bad ideas from being implemented.

2.2 Contribution to organizational economics

The paper provides a formal comparison of two architectures:

- **Polyarchy:** decision makers operate in parallel; a project can be accepted if at least one unit accepts it.
- **Hierarchy:** decision makers operate in series; a project is accepted only if it passes all levels of review.

This distinction allows the authors to map organizational form into acceptance probabilities, final portfolio size, expected output, and error composition.

2.3 Conceptual contribution

The paper's most useful conceptual move is to translate organizational design into a statistical decision problem. Individual screens make Type-I errors, rejecting good projects, and Type-II errors, accepting bad projects. The architecture then determines whether these errors are compounded, corrected, or amplified.

3 Economic environment

3.1 Projects

The organization faces a set of projects. Each project has net benefit

$$x,$$

which may be positive or negative. Projects with $x > 0$ are socially or economically desirable, while projects with $x < 0$ should ideally be rejected.

The distribution of projects is denoted by

$$g(x),$$

and there are N available projects. The number N is often normalized away, so performance is discussed per project or as a proportion of the initial portfolio.

3.2 Screening

Each individual evaluator observes a project imperfectly. The screening function is

$$p(x),$$

where $p(x)$ is the probability that a project of quality x is judged good and accepted by an individual screen.

The authors assume

$$p_x(x) > 0,$$

so higher-quality projects are more likely to be accepted. However, screening is imperfect: a good project can be rejected and a bad project can be accepted.

3.3 Two dimensions of a screen

The paper emphasizes two properties of screening.

Key properties of the screening function

Property	Meaning	Economic interpretation
Slackness, $p(x)$	Higher level of acceptance probability at a given project quality	A slacker screen accepts more projects, including both good and bad ones.
Discriminating ability, $p_x(x)$	Steeper relation between quality and acceptance probability	A more discriminating screen is better at separating high-quality from low-quality projects.

A useful example is a linear screening function:

$$p(x) = p(\mu) + p_x(x - \mu),$$

where $\mu = E[x]$ is the mean quality of the initial project portfolio.

4 Two architectures: polyarchy and hierarchy

4.1 Polyarchy

A polyarchy is a system in which multiple decision makers can accept projects independently. In the paper's two-unit version, a project is first reviewed by one firm. If the first firm accepts it, the project enters the final portfolio. If the first firm rejects it, the project can still be reviewed by the second firm. Thus, a rejected project receives a second chance.

If each firm uses the same screening function $p(x)$, the probability that a project is accepted by the polyarchy is

$$f^P(x) = p(x) + (1 - p(x))p(x) = p(x)(2 - p(x)).$$

4.2 Hierarchy

A hierarchy is a system in which projects must pass sequential review. In the two-unit version, a project is first reviewed by a lower bureau. Only projects approved by the lower bureau are sent to the higher bureau. A project is implemented only if both bureaus accept it.

If both bureaus use the same screening function $p(x)$, the probability that a project is accepted by the hierarchy is

$$f^H(x) = p(x)^2.$$

4.3 Core mechanical comparison

For every x with $0 < p(x) < 1$,

$$f^P(x) - f^H(x) = p(x)(2 - p(x)) - p(x)^2 = 2p(x)(1 - p(x)) > 0.$$

Therefore, a polyarchy always accepts a larger proportion of projects than a hierarchy, holding fixed the individual screening function.

Acceptance probabilities under the two architectures

Architecture	Acceptance probability	Intuition
Polyarchy	$f^P(x) = p(x)(2 - p(x))$	Parallel review; one acceptance is enough.
Hierarchy	$f^H(x) = p(x)^2$	Sequential review; both levels must accept.
Difference	$f^P(x) - f^H(x) = 2p(x)(1 - p(x))$	Polyarchy accepts more projects whenever screening is imperfect.

5 Main theoretical results

5.1 Result 1: polyarchy selects more projects

Let

$$n^s = E[f^s(x)]$$

denote the proportion of the initial project portfolio selected by system s , where $s = P$ for polyarchy and $s = H$ for hierarchy. Because $f^P(x) > f^H(x)$ for imperfect screens,

$$n^P > n^H.$$

Economic message: A polyarchy has a larger final portfolio. It accepts more good projects, but also more bad projects. A hierarchy has a smaller final portfolio. It rejects more bad projects, but also more good projects.

5.2 Result 2: error composition differs across architectures

The architecture changes the relative incidence of Type-I and Type-II errors.

System-level errors

Error type	Definition	More prevalent in
Type-I error	Rejecting a good project	Hierarchy
Type-II error	Accepting a bad project	Polyarchy

Economic message: Polyarchy protects against missed opportunities; hierarchy protects against bad approvals.

5.3 Result 3: initial portfolio quality matters

When the available project pool improves, both systems accept more projects. But the relative performance of the systems depends on whether the project pool contains many good ideas or many bad ideas.

If the project pool is bad, the hierarchy's advantage in rejecting bad projects becomes more valuable. If the project pool is good, the polyarchy's advantage in rescuing good projects becomes more valuable.

5.4 Result 4: mean-preserving spreads can affect systems differently

With a linear screening function, a mean-preserving spread in the initial project distribution has opposite effects on the two architectures:

- It decreases the proportion selected in a polyarchy.
- It increases the proportion selected in a hierarchy.

This occurs because the polyarchy's acceptance function is concave in the screening probability, while the hierarchy's acceptance function is convex.

5.5 Result 5: slackness and discrimination have different effects

With a linear screening function:

- Higher slackness raises the proportion selected in both systems.
- Higher discriminating ability lowers the proportion selected in a polyarchy but raises it in a hierarchy.

The second result is less obvious. Greater discrimination makes acceptance probabilities more sensitive to quality. Because the two architectures transform $p(x)$ differently, increased discrimination changes aggregate selection differently.

6 Expected output with two types of projects

The paper then studies a simple project pool with only two project types:

- Good projects yield $z_1 > 0$.
- Bad projects yield $-z_2 < 0$.
- The fraction of good projects is α .

Let

$$p_1 = p(z_1), \quad p_2 = p(-z_2),$$

where p_1 is the probability of accepting a good project and p_2 is the probability of accepting a bad project. Since the screen is informative, $p_1 > p_2$.

The difference in expected output between polyarchy and hierarchy is

$$\Delta Y = Y^P - Y^H = 2z_2(1 - \alpha) [ap_1(1 - p_1) - p_2(1 - p_2)],$$

where

$$a = \frac{z_1\alpha}{z_2(1 - \alpha)}$$

measures the quality of the initial portfolio.

6.1 Interpretation of the output comparison

The term $p_1(1 - p_1)$ is related to the probability that a good project receives mixed evaluations, while $p_2(1 - p_2)$ is related to the probability that a bad project receives mixed evaluations. A polyarchy accepts mixed-review projects; a hierarchy rejects them. Therefore, the sign of ΔY depends on whether mixed-review projects are, on average, valuable or harmful.

6.2 Moderately good portfolio: $a = 1$

When $a = 1$, a polyarchy performs better if

$$1 - p_1 > p_2.$$

Here $1 - p_1$ is the individual screen's Type-I error rate, and p_2 is the Type-II error rate. If the screen is more likely to wrongly reject a good project than wrongly accept a bad project, then the second chance provided by polyarchy is especially valuable.

When does each architecture do better?

Condition	Preferred architecture
Good projects are often wrongly rejected	Polyarchy, because it gives projects another chance.
Bad projects are often wrongly accepted	Hierarchy, because it requires multiple approvals.
Initial project pool is strong	Polyarchy becomes more attractive.
Initial project pool is weak	Hierarchy becomes more attractive.
Communication is costly or noisy	Polyarchy becomes more attractive because it requires less coordination.

7 Endogenous screening rules

The paper then moves beyond exogenous screening functions and asks how agents choose their screening rules.

7.1 Signal structure

An evaluator observes a noisy signal

$$y = x + \theta,$$

where x is project quality and θ is noise. The evaluator uses a reservation rule: accept the project if

$$y \geq R.$$

The resulting screening function is

$$p(x, R) = \Pr[y \geq R] = 1 - M(R - x),$$

where M is the distribution function of the noise term.

A higher reservation level R makes the screen tighter:

- It increases the probability of rejecting good projects.
- It decreases the probability of accepting bad projects.

7.2 Reservation levels differ by architecture

Let R^H be the optimal reservation level in a hierarchy, R^P the symmetric Nash reservation level in an uncoordinated polyarchy, and R^C the reservation level in a coordinated polyarchy. The paper shows:

$$R^C > R^P > R^H.$$

Economic message: Screening in a polyarchy is more conservative than in a hierarchy, because a firm in a polyarchy knows that projects it receives may already have been rejected elsewhere and that its own acceptance will not be rechecked. A coordinated polyarchy is even more conservative because it internalizes the negative externality that one firm imposes on another when it rejects projects into the other's portfolio.

7.3 How portfolio quality affects reservation levels

If the project pool contains more bad projects, both systems tighten their screens. Formally, a worsening of the available portfolio increases reservation levels under hierarchy, uncoordinated polyarchy, and coordinated polyarchy.

Economic message: Organizations should become more selective when the environment contains more low-quality opportunities.

8 Applications and interpretation

8.1 Markets versus bureaucracies

The paper interprets polyarchies as capturing some features of market-oriented systems and hierarchies as capturing some features of bureaucracy-oriented systems.

In a market-like polyarchy, independent firms can pursue ideas separately. If one firm rejects an idea, another firm may still implement it. This reduces missed opportunities but increases the possibility that bad projects are undertaken.

In a bureaucracy-like hierarchy, projects undergo sequential approval. This reduces the chance that bad projects are implemented, but it also increases the chance that good ideas are killed by the process.

8.2 Communication costs

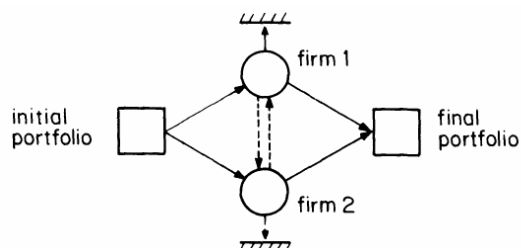
A central assumption is limited communication. In a hierarchy, information must pass from one level to another, but this communication is imperfect and often binary: good or bad, pass or reject. In a polyarchy, firms may not communicate at all, which can be an advantage when communication is costly or noisy.

8.3 Relation to innovation

The model is especially relevant for innovation. Innovative projects are uncertain and difficult to evaluate. If good ideas are rare but extremely valuable, a polyarchy may be attractive because it gives ideas multiple chances. If bad ideas are abundant and costly, a hierarchy may be attractive because it creates stronger filters.

9 Extracted figures from the paper

9.1 Figure 1: Polyarchy and Figure 2: Hierarchy



(a) Figure 1. Polyarchy

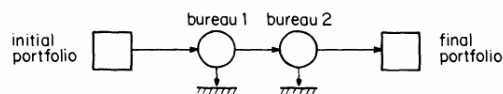


FIGURE 2. HIERARCHY

(b) Figure 2. Hierarchy

Read:

- Figure 1 depicts a polyarchy: projects move from the initial portfolio to one firm and, if rejected, may be reviewed by another firm.
- Figure 2 depicts a hierarchy: projects must pass from bureau 1 to bureau 2 before entering the final portfolio.
- The visual distinction is parallel versus serial screening.

Economic message: The same individual screening function generates different aggregate decision rules. In the polyarchy, one approval is sufficient; in the hierarchy, two approvals are required.

9.2 Figure 3: Probabilities of Acceptance in Alternative Systems and Figure 4: Comparison of a Polyarchy and a Hierarchy

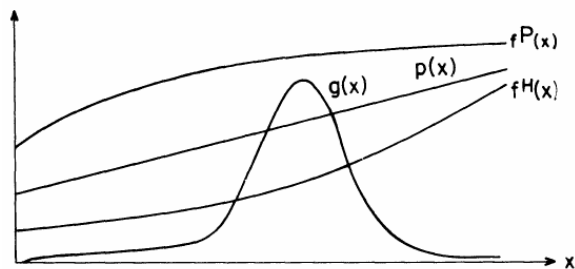


FIGURE 3. PROBABILITIES OF ACCEPTANCE IN ALTERNATIVE SYSTEMS

(a) Figure 3. Probabilities of Acceptance in Alternative Systems

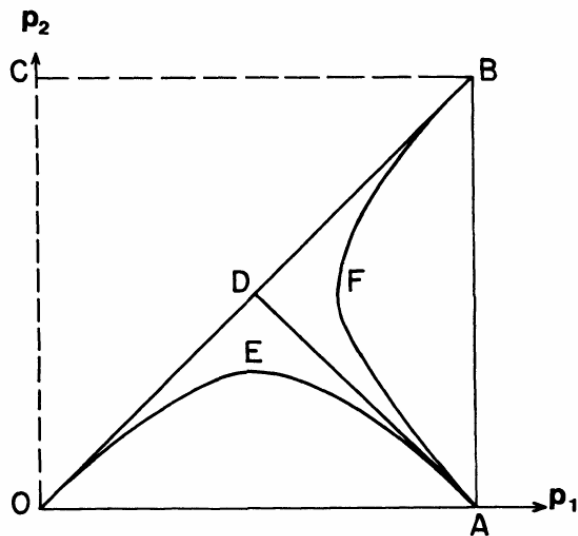


FIGURE 4. COMPARISON OF A POLYARCHY AND A HIERARCHY

(b) Figure 4. Comparison of a Polyarchy and a Hierarchy

Read:

- Figure 3 plots the individual screening function $p(x)$ and the aggregate acceptance functions $f^P(x)$ and $f^H(x)$.
- The polyarchy acceptance probability lies above the hierarchy acceptance probability because $p(2 - p) > p^2$ for imperfect screens.
- Figure 4 divides the (p_1, p_2) parameter space into regions where polyarchy or hierarchy performs better.

Economic message: Figure 3 shows why polyarchies select larger final portfolios. Figure 4 shows that larger selection is not always better: the preferred architecture depends on the relative probabilities of rejecting good projects and accepting bad projects, as well as the quality of the initial project pool.

10 How to use this paper in research

10.1 Useful mechanism

The paper provides a general mechanism for research on organizations, finance, innovation, and policy institutions:

When agents make noisy judgments, institutional design matters because it determines whether mistakes are corrected, compounded, or transmitted.

10.2 Possible empirical analogues

The model suggests empirical questions such as:

- Do decentralized funding systems accept more risky projects than centralized grant committees?
- Do multi-layer approval processes reduce bad investments but also suppress innovation?
- Are decentralized markets better when project quality is difficult to assess ex ante?
- Do organizations become more hierarchical when the cost of false positives is high?
- Do organizations become more polyarchical when the cost of false negatives is high?

10.3 Connection to finance

In finance, the paper helps interpret venture capital, bank lending, investment committees, and capital allocation inside firms. A venture capital ecosystem resembles a polyarchy: a founder rejected by one investor can approach another. A bank credit committee resembles a hierarchy: loans may require multiple approvals. The model predicts that VC-style systems may fund more bad projects but also discover more outstanding projects, while bank-style systems may avoid losses but miss valuable opportunities.

11 Short summary

- The paper studies how alternative economic architectures aggregate imperfect project evaluations.
- A polyarchy places screens in parallel: a project can be accepted if any unit accepts it.
- A hierarchy places screens in series: a project must be accepted by all levels.
- With identical individual screening, a polyarchy accepts a larger proportion of both good and bad projects.
- Therefore, hierarchy has more Type-I errors, rejecting good projects, while polyarchy has more Type-II errors, accepting bad projects.
- Expected performance depends on the project pool, the relative cost of the two errors, and the accuracy and tightness of screening.
- When screening rules are endogenous, polyarchies choose tighter reservation levels than hierarchies.
- Communication costs and limited information are central to why architecture matters.

12 Conclusion

Sah and Stiglitz (1986) offer a compact but powerful theory of organizational architecture. The paper's main insight is that markets and bureaucracies differ not only in incentives or resource allocation, but also in how they process mistakes. Polyarchies give good ideas more chances but also allow more bad ideas through. Hierarchies impose stronger checks but also kill more good ideas. The best architecture depends on the environment: when missed opportunities are costly, decentralization is valuable; when bad approvals are costly, hierarchy is valuable.